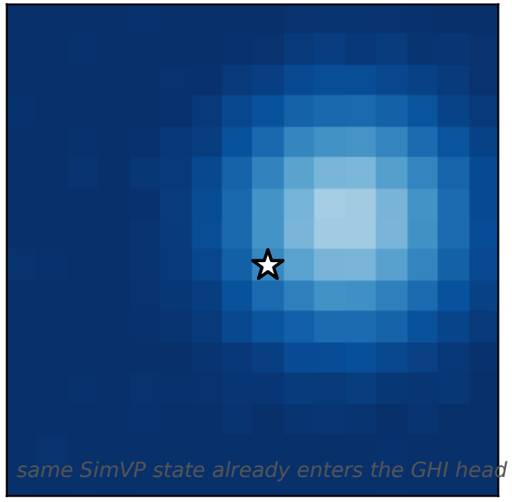
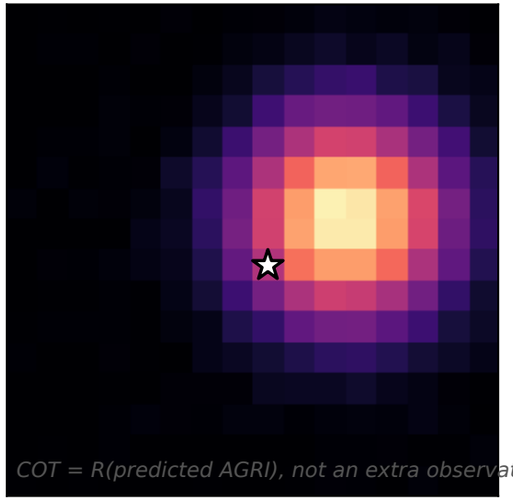
# Why a predicted COT map at every future time may still show no GHI gain
# Illustrative mechanisms and planned tests; this figure contains no experiment values.

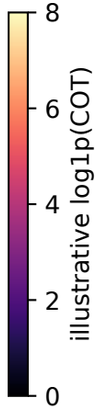
Predicted AGRI at target time




frozen R

Predicted COT at the same target time



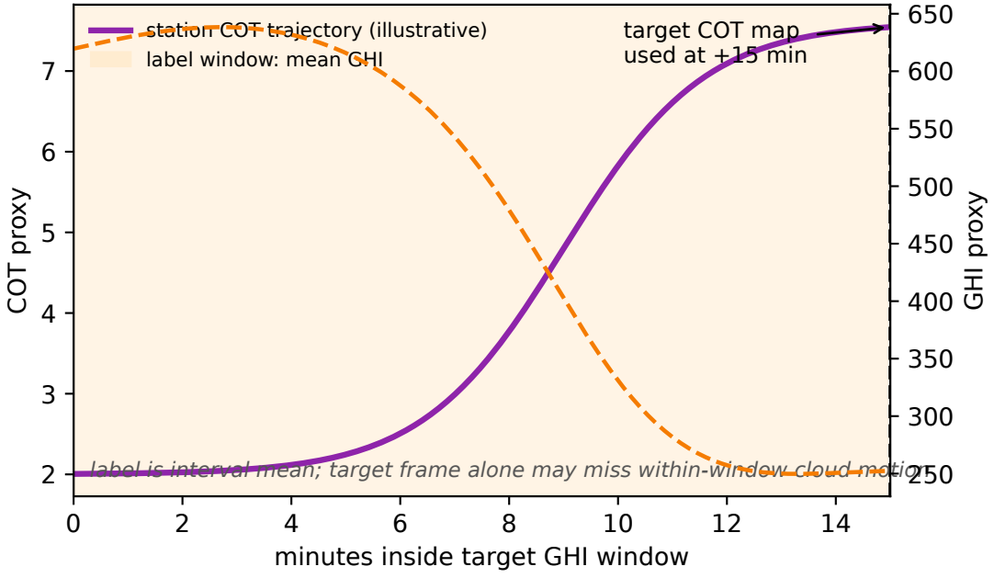

## A target-time COT map can help

when R makes cloud optical thickness easier for a finite GHI head to use.

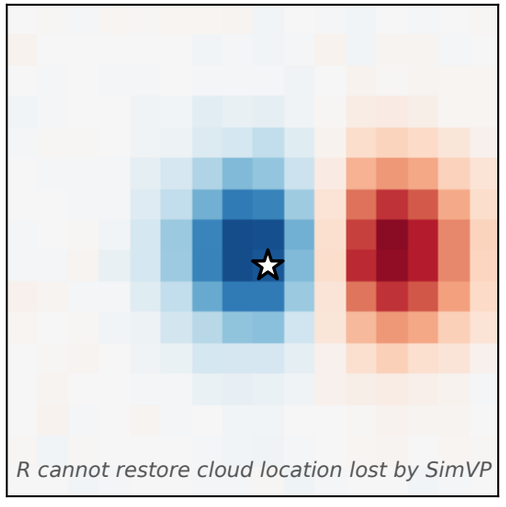
It has no guaranteed new information: COT is a deterministic re-expression of the same predicted AGRI field.

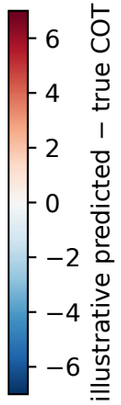
Thus an AC_F−A_ST gain is an empirical representation result, not automatic.

A single COT frame is an endpoint snapshot




Forecast-position error becomes COT-position error




## What the paired ablation resolves

**AC_F − A_ST**

Does the same-time predicted COT map help?

**AC_ST − AC_F**

Does history + causal prefix add trajectory value?

**COT retrieval audit**

Is R accurate enough on forecast AGRI?

**per-lead / cloud regime**

Is any gain confined to moving thick cloud?